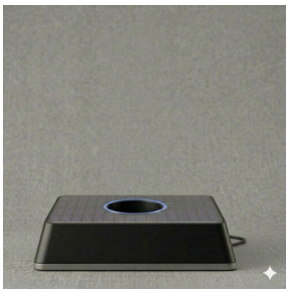
Design Milestones

IDE202.A

Gülçe Gez
S045812

Timeline



CIRCULAR COFFEE GRINDER HOPPER
FLAWLESS BRUSHED METAL
BRUTHED MATHE FINISH
SOLAR-POWERED GRINDER
DOORS WITH REFINED HANDLES

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Emphasize

AI's Role: Clarifier

AI's Tools Used: Gemini

INPUT

Prompt:

What are things that people look for in tea/coffee making machines? Can you find some information through surveys and researches?

Attachments: None

OUTPUT

AI Text-Summary:

A detailed analysis highlights that consumers prioritize not just the convenience and quality of their tea and coffee machines, but also their ability to offer innovative features, design appeal, and variety in brewing capabilities. Understanding these trends is crucial for manufacturers to adapt offerings to meet the demands of modern consumers.

This comprehensive overview captures the essential aspects influencing consumer choices in the tea and coffee machine market, reflecting ongoing trends and shifts driven by lifestyle preferences.

Core Consumer Priorities

Quality & Customization: Users demand consistent flavor and aroma, but they also want granular control over variables like temperature, brewing time, and strength to personalize their beverages.

Operational Convenience: A "one-touch" interface is highly valued. Equally important is the ease of maintenance, with a strong preference for removable parts and automated cleaning cycles.

Versatility: Consumers are gravitating toward "hybrid" machines that can handle multiple drink types (coffee, tea, and specialty beverages) with pre-set programs for efficiency.

Modern Aesthetics & Footprint: A sleek design that matches kitchen décor is a major influencer, specifically in compact forms that save counter space.

Evolving Market Trends

The Home-Brewing Shift: Since the pandemic, nearly 45% of coffee drinkers have adopted new brewing methods at home, increasing the demand for high-end residential equipment.

Sustainability & Ethics: There is a growing focus on energy efficiency and recyclable materials, particularly among younger demographics like Gen Z, who value ethical sourcing and "café-style" novelty.

Connected Technology: Integration with mobile apps for scheduling, maintenance alerts, and remote brewing is becoming a standard expectation for the modern "smart home."

Decisions: With these informations I've learned some of the issues within the current electronic hot beverage makers and decided to focus on some issues seperately.

Define

AI's Role: Clarifier
AI Tools Used: Claude/Gemini

INPUT

Prompt:

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Can you tell me about coffee machines used in caravans/boats
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Attachments: None

OUTPUT

AI Text-Summary: Coffee machines for caravans and boats must handle limited power, space, and water. They generally fall into three categories:

1. No-Power Options
These are best for off-grid use and only require hot water or a stove.
Moka Pot – simple stovetop brewer that makes strong coffee.
French Press – compact and easy, needs only hot water.
AeroPress – lightweight, durable, and popular with travelers.
Pour-over – simple cone and filter system.
Manual espresso makers like Wacaco Nanopresso or Flair Espresso Maker produce espresso without electricity.
2. 12V / Low-Power Machines
Made for vehicles and boats.
12V espresso machines such as Handpresso models plug into a car socket.
12V pod machines using capsule systems (like Nespresso) offer convenience but create pod waste.
3. Mains-Powered Machines (when connected to electricity)
When plugged into campsite power or a strong inverter:
Pod machines like Nescafé Dolce Gusto or Nespresso
Filter coffee machines
Bean-to-cup machines (higher power use)
Key considerations:
Wattage (power limits in caravans)
Water usage
Size and weight
Stability, especially on boats
Most practical choices: the AeroPress and Moka Pot because they are simple, compact, and don’t require electricity.

INPUT

Prompt: Can you tell me what the users expect/want?

Attachments: None

OUTPUT

AI Text-Summary: In short, the ideal machine for this audience is simple, tough, compact, low-power, easy to clean, and makes great coffee. The fact that no single electric machine perfectly ticks all those boxes is exactly why manual options remain so dominant in this space. Would you like to explore any of these areas further?

Decisions: Before this with some research I've decided my user and I chose caravan/boat users. With these questions I understood my user better and issues that can be looked at. I've decided to pick up on the issues like being able to use in various situations and safety. Furthermore I ended up changing my user to people who use caravans/campers.

- Reliable
- Safe
- Energy-Sufficient
- Compact
- Multi-functional

Ideate

AI's Role: Expander/Render

AI Tools Used: Vizcom

Early Design Concepts:

Reflection: I noticed that instead of giving it the attributes of a "counter" type of coffee machine and excepting it to do everything I should focus on it being compact and still up to the standards of my selected user.

Decisions: I've decided to not use these concepts but instead make something that is more compact and more usable in the outdoors. Also I wanted to integrate solar power into the machine to make the machine more usable in different scenarios.

AI's Role: Critique

AI Tools Used: Gemini

INPUT

Prompt:

```
Would a coffee machine like this could work? The batteries are 12V.

Would it run better with two 12V batteries? And would everything that is needed
mechanically fit into the machine?

I need to fit all the mechanical parts in a 120mm tall 90mm diameter space is that
possible?
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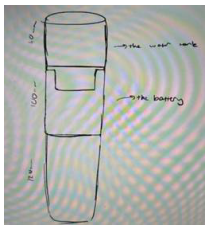
Attachments:



OUTPUT

AI Text-Summary: Technical Constraints to Watch

• Heat Dissipation: In a 120mm enclosed space, the pump and the base of the water heater will generate a lot of heat. Using Polypropylene for the internal structural frame is a smart move because it’s a good thermal insulator and won't melt under standard brewing temperatures.

• The Seal: Your "bottom lock" must be able to withstand roughly 130 psi (9 bar). In a 90mm diameter space, a bayonet-style twist lock (similar to a camera lens mount) is usually the most space-efficient way to keep the machine from leaking under pressure.

Summary

Yes, it fits, provided you use micro-components rather than scaled-down versions of home kitchen parts. The 90mm width is your best friend here—it gives you enough volume to "nest" the pump alongside the brew chamber rather than stacking them, which saves you a lot of height.

Reflection: With the information AI provided I learned that 2 12V batteries are possible but it can cause some issues if not done properly. One of the biggest energy consumer is the pump so AI suggested that I get rid of the pump and do a mechanism like a french-press. Since I want the machine automatic I will not go with that.

Decisions: I've decided to change the place of the place of the ground coffee and put it in between the mechanical parts and the cup, that way because of the gravity the pump needed won't be that strong making the coffee machine more stable. As an addition I've decided to make a separate charging station and a coffee grinder based on the previous suggestions and issues that AI mentioned.

Prototype/Test

- Reliable
- Safe
- Energy-Sufficient
- Compact

Main Design Concept

AI's Role: Render
AI Tools Used: Gemini

AI's Role: Clarifier/ Reframe-er
AI Tools Used: Gemini

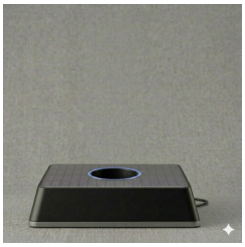
OUTPUT
AI Text-Summary: The design is a uni body 90mm cylinder that focuses on a modular "Power-Cup" concept. In that iteration, the 6-cell lithium-ion battery pack was integrated into the base of the cup itself, serving as a detachable power source that would "mate" with the heater via electrical contacts when docked. While this kept the overall profile perfectly flush and sleek at 28cm tall, it presented significant engineering hurdles: the cup was heavy and non-submersible, and the high-current connection between the cup-batteries and the top-mounted heater was highly vulnerable to "arcing," dirt, and moisture—common issues in a caravan environment.

Reflection: With the information AI provided I drafted away from the coffee machine charging from cup because of the mechanical difficulties.

Decisions: I've decided to make a separate cup so you can charge it without the cup and the mechanical issues are addressed accordingly. And with that I had to change the place where you put the coffee grounds too. So I decided to make a dent inside for the coffee. And I've decided to integrate solar power into it as a second way of use.

Main Design Concept

AI's Role: Render
AI Tools Used: Gemini/Chat-GPT



Reflection: I've realized that while trying it to give it a bigger base for faster solar charging made me draft away from making the product compact.

Decisions: I've decided to get rid of the square base and explore more options on how can I cover less ground but still have a good enough area for the solar panels.

Prototype/Test

Final Design

AI's Role: Render
AI Tools Used: Gemini

Front (Closed)

Front (Half-Open)

Back

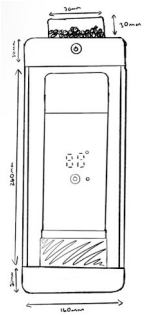
Decisions: I've decided that this design will benefit the user most.

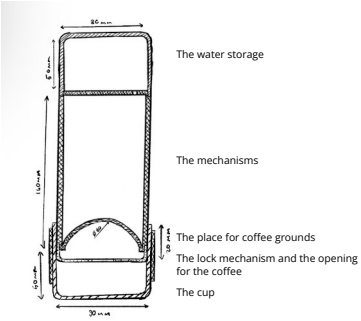
- Since caravans have windows all around the sunlight can always enter the inside of the caravan the three solar panels that can surround the machine allows the machine to get sunlight the whole day. The two panels at the front open with a sliding motion while the one on the back is stable.
- When the user wants to charge the coffee machine all they need to do is twist the cup out of its place and twist it onto the "charging station" that works with the bayonet twist-lock mechanism.
- The user can grind their own beans at the top of the machine.
- The solar panels add an additional way of charging the machine when the user does not want to use the accumulator/generator for the machine.
- The machine can also self-clean by pressing the button on the middle right.

Polypropylene (PP): For the body

Thermoplastic Elastomer (TPE): For the grip part

Stainless Steel (304or316 Grade): For the details





The lock mechanism that will be used between the "charging station" and the coffee machine(just a smaller version)

AI's Role: Critique
AI Tools Used: Gemini

OUTPUT
AI Text-Summary:

Sunlight Quality	Estimated Power Output	Time to Full Charge (0-100%)
Peak Direct Sun	12W – 15W	2.5 to 3.5 Hours
Partial Sun/Haze	5W – 7W	5 to 7 Hours
Overcast/Indirect Light	1W – 2W	15+ Hours (Multiple days)